W. Shakespeare

Categories and functors: much ado about nothing?

Bachelor thesis

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Thesis supervisors: **prof.dr. G.A. Bredero**
dr. J. van den Vondel

Leiden University
Mathematical Institute

Contents

1 Introduction

1.1 Assumptions

The following things are assumed to be true

1. The group of people that the model is made for is very small compared to daily travelers,therefore this small group will have no effect on overcrowdedness or other usage related metrics.

2 Algorithm

This section summarizes the algorithm developed by Rumpf and Kaul in [RK21]. As the model and algorithm is made by them, a more detailed explanation can be found in their paper. In this section we will give a summary so the later parts of this study can be understood.

2.1 SAMP

The algorithm made by Rumpf and Kaul in [RK21] was made to solve the *social access maximization problem* (SAMP). The formulation of the problem is very general as to be aplicable to many situations.

Let L be the set of considered transitlines, where a transitline $l \in L$ is a series of stops with a given traveltime between each stop. A schedule y is a vector with $y_l \in \mathbb{N}$ the amount of vehicles assigned to line l in a given time period. The vehicles are distributed evenly over the time period, so 5 vehicles in an hour gives 1 trip every 12 minutes.

The initial schedule is denoted as y^*. The model has upper and lower bounds $y_l^{\min} \leq y_l \leq y_l^{\max}$ for all $l \in L$. These can be set to 0 and infinity if no ristriction is needed, or both to y_l^* if no change is wanted.

Let x be the user flow vector for the transit network, x^* the initial flow vecor. User flow represents how busy parts of the network are, which is influenced by the schedule y and found by the TransitAssignment(y) function, which will be explained further in Section 2.2.

Other restraints of the model are given by operating- and user costs with upper bounds $B_{operator}, B_{user}$ respectively. Operating costs are mostly made up by monetary costs of running the transit network and monetary gain of users paying to use the service. User costs are made up of difficulty in traversing the network, be that in monetary cost or time spent in the system. More on these costs in Sections 2.3.

Let Access(y) be a function describing the social access objective. This value only depends on the design of the network y and not on user flow x, because the model is designed for a small group of individuals following assumption 1 Different choices for Access(y) are given in 2.4.

2.2 Transit Assignment

2.3 Costs

2.4 Access

2.5 The model

The SAMP model then takes the shape of the following nonlinear program

$$\max_y \quad \mathrm{Access}(y) \tag{1}$$

$$\text{s.t.} \quad \sum_{l \in L} y_l \leq \sum_{l \in L} y_l^* \tag{2}$$

$$y_l^{\min} \leq y_l \leq y_l^{\max} \quad \forall l \in L \tag{3}$$

$$y_l \in \mathbb{N} \qquad \forall l \in L \tag{4}$$

$$x = \mathrm{TransitAssignment}(y) \tag{5}$$

$$\mathrm{OperatorCost}(y, x) \leq (1 + \varepsilon)\mathrm{OperatorCost}(y^*, x^*) \tag{6}$$

$$\mathrm{UserCost}(x) \leq (1 + \varepsilon)\mathrm{UserCost}(x^*) \tag{7}$$

2.6 TODO

The algortithm was designed to solve a general **Social Access Maximization Problem** or SAMP, given by

3 Acquiring data

To run the algorithm, some data needs to be acquired. This section will provide the methodology of how this data was acquired, full tables of data can be found in the appendix.

3.1 Busses

3.1.1 Buslines

Only buslines that travel at least partially through the study area are taken into account for this study. Lines that are not entirely inside the study area are truncated to only have stops within it. The stop were such lines enter or leave the municipality are treated as starting- or ending stops, respectivily. This gives that some lines that differ outside the municipality appear identical in our study.

3.1.2 Busschedule

After the buslines were selected, a schedule was retreived from [Qbu25]. The chosen schedule was for monday the 10th of february of 2025, between 14:00 and 15:00 in the afternoon. This weekday and timeframe was chosen as a ordinary, non-rush hour time, that represents ordinary bus travel.

The infromation given by a schedule is average traveltime between stops, wait time at a stop and amount of busses per line per hour. The algorithm used takes busses per line per timeframe and averages this out to arrive at the frequency of a line, where more busses per timeframe would mean a higher frequency of service.

3.1.3 Busstops

Traveltime between busstops was extracted from the schedule, as was their name. The other information needed was their location, so that distances between origin or destination points can be found and walking arcs could be connected between busstops, GP offices and community centers. As most busstops consist of a pair of physical locations, one for each direction, their midpoint is taken as the location of the pair overall.

3.2 Destination points

The destination points are given by the location of general practitioners (GP's) in Leiden. The GP-offices have a location and a quality metric. A full list of GP's in Leiden can be found on [Ned25], including a link for each individual office's website. This also gives the location and amount of full-time employed doctors. The latter was taken as the quality of a GP-office, where more doctors would imply a higher capacity of patients and thus a higher quality.

3.3 Population

All trips originate from a community center. Each community center is a point, representing a region. All trips within a region are assumed to originate from that regions community center. From this center, walking arcs are made connecting nearby bus stops. Each community center has a location and a population. The location determines which busstops can be connected by a walking arc and the population is used by the algorithm to determine the amount of trips originating from a community. Two methods of dividing the municipality into regions are used and compared, namely the PC4 regions and voronoi regions.

3.3.1 PC4

Each adress in the Netherlands has a postcode, consisting of four numbers and two letters. If one only uses the four numbers, the Netherlands can be divided into pc4 regions. This division roughly resembles which areas were built at the same time.

The location of each community center is taken as the rough middlepoint of the region, and the population is taken from [All25].

3.3.2 Voronoi regions

According to [Wan22], travelers prefer to take a bus at the nearest bus stop, even when that involves a transfer instead of a direct route. To apply this effect on our study, the regions we choose should reflect trips that start at the closest bus stop. A voronoi diagram is the ideal method for this.

A voronoi diagram is created by using a set of weightpoints in a plane, we use the busstops as weightpoints. Via an algorithm, the plane is divided in polygons. Each polygon contains exactly one weightpoint and all points in a polygon have that weightpoint as their closest weightpoint.

When the plane is infinite, the outer voronoi regions will also be infinite. For this study, a border had to be drawn to limit all voronoi regions to be within the municipality of Leiden.

We again need a community center location and a population. Since each bus stop is taken as a weightpoint, these also serve as the community center location. To approximate the population of each voronoi region we estimate the overlap between a voronoi- and a pc4 region. Let

$$\begin{aligned} V &\text{ set of voronoi regions,} \\ P &\text{ set of postcode regions,} \\ s_v &\text{ surface area for } v \in V, \\ d_p &\text{ population density for } p \in P, \\ f(v,p) &\text{ fraction of } v \text{ that lies in } p. \end{aligned}$$

Then, the population of a voronoi region v is given by

$$s_v \sum_{p \in P} d_p f(v,p).$$

4 Results

5 Discussion

References

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